Project Report

Heart Attack Prediction System Using Machine Learning

Abstract:

Heart attacks are among the most life-threatening cardiovascular events and often occur without early visible symptoms. Predicting the likelihood of a heart attack at an early stage can significantly reduce mortality rates. This project presents a **Heart Attack Prediction System** developed using machine learning techniques. The system is trained on the **Cleveland Heart Disease Dataset**, which contains real patient clinical data related to cardiovascular health. A **Random Forest classifier with 300 decision trees and a maximum depth of 10** was employed to learn complex patterns associated with heart attack risk. The trained model was saved using **joblib** as `heart_attack.pkl` and deployed through an **interactive Streamlit web application**. The system allows users to enter medical parameters and receive real-time predictions along with probability scores, making it an effective educational and preliminary risk assessment tool.

Problem Statement:

Heart attacks are a major cause of death worldwide, and delayed diagnosis is one of the primary reasons for high fatality rates. Many individuals do not have immediate access to medical experts or advanced diagnostic facilities. The challenge addressed in this project is to **design an automated system that predicts the risk of a heart attack using common clinical indicators**, enabling early awareness and timely medical consultation.

Introduction:

A heart attack occurs when blood flow to the heart muscle is blocked, often due to coronary artery disease. Factors such as high blood pressure, cholesterol levels, abnormal ECG results, and exercise-induced angina significantly increase the risk of heart attacks. Traditional diagnosis relies on medical tests and expert interpretation, which can be expensive and time-consuming.

With the advancement of machine learning, historical medical datasets can be analyzed to identify patterns associated with heart attacks. This project applies machine learning techniques to predict heart attack risk based on patient data. The **Cleveland Heart Disease Dataset** was chosen because it is widely used in medical research and contains clinically relevant attributes. The ultimate goal is to develop a **reliable and interactive heart attack prediction system** that assists in early risk identification.

Methodology:

1. Dataset Description:

- Dataset: Cleveland Heart Disease Dataset

- Number of records: 303
- Number of features: 13 clinical attributes
- Target variable:

    - `0` → No Heart Attack Risk
    - `1` → High Risk of Heart Attack

2. Data Preprocessing:

- Missing and inconsistent values were handled
- Target labels were converted into binary format
- Categorical medical attributes such as chest pain type, sex, ECG results, and thalassemia were encoded numerically
- Feature order was fixed to ensure consistency between training and deployment

3. Model Training:

- Algorithm used: **Random Forest Classifier**
- Hyperparameters:

    1. Number of trees (`n_estimators`): 300
    2. Maximum tree depth (`max_depth`): 10
    3. Class balancing applied to handle class imbalance

- The dataset was split into training (80%) and testing (20%) subsets

4. Model Saving:

- The trained model and feature information were saved using **joblib**
- File name: `heart_attack.pkl`
- This ensured reliable reuse of the model during deployment

5. Deployment:

- An interactive **Streamlit web application** was developed
- Users can input patient clinical values through sliders and dropdown menus
- The system outputs:

    1. Heart attack risk classification
    2. Probability percentage of heart attack
    3. Visual probability bar chart

Results:

The trained Random Forest model achieved strong predictive performance:

- **Training Accuracy:** ~87%
- **Testing Accuracy:** ~85–90%

Prediction Examples:

- **Low-Risk Case:**

  1. Young patient, normal blood pressure, normal cholesterol
  2. Predicted heart attack probability: ~15%

- **High-Risk Case:**

  1. Older patient, high cholesterol, exercise-induced angina
  2. Predicted heart attack probability: ~90%

The Streamlit application successfully provides real-time predictions and visual feedback, improving user understanding of heart attack risk.

Discussion:

The Random Forest model proved effective for heart attack prediction due to its ability to handle nonlinear relationships and multiple clinical features. The high number of decision trees increased stability and reduced overfitting. Saving the model using joblib ensured smooth integration with the web application.

While the system provides accurate predictions, it is intended for **educational and preliminary assessment purposes only**. The dataset size and demographic limitations may affect generalization. Future improvements could include integrating additional datasets, explainable AI techniques, and real-time hospital data.

Conclusion:

This project successfully demonstrates a **Heart Attack Prediction System** using machine learning. By training a Random Forest classifier on the Cleveland Heart Disease Dataset and deploying it through an interactive Streamlit application, the system enables early identification of heart attack risk. The project highlights the potential of machine learning in healthcare and provides a strong foundation for future medical decision-support systems.

Video Link:

https://drive.google.com/file/d/1acongo9I05mDfyHTa2jKUf6y8Q3bSCdH/view?usp=sharing